

Pythagoras' theorem

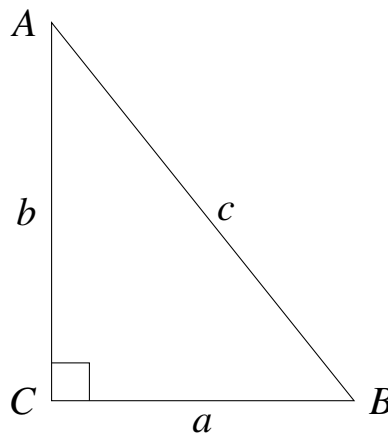
Covering

- Pythagoras' theorem

Pythagoras' theorem

The Pythagoras theorem is a fundamental principle in geometry that relates the sides of a right-angled triangle.

For a triangle ABC , with angle $C = 90^\circ$,



a and b are the legs, the shorter sides, and c is the hypotenuse, which is opposite to the right angle and is the longest side.

The formula that relates the three sides is $a^2 + b^2 = c^2$, where a and b are the lengths of the sides of the right triangle, and c is the length of the hypotenuse.

We can solve for each of the variables by rearranging them.

$$a = \sqrt{c^2 - b^2} \quad b = \sqrt{c^2 - a^2} \quad c = \sqrt{a^2 + b^2}$$

The basic integer solution for a , b and c is commonly $3^2 + 4^2 = 5^2$.



Trigonometry

Covering

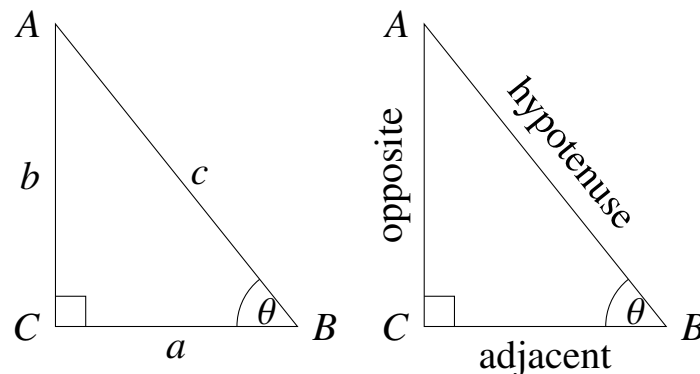
- Right-angled triangles
- Non-right-angled triangles
- Pythagoras' theorem and trigonometry in 3D

O levels and Extended

O levels and Extended

Basic Ratios

For a triangle ABC , with angle $C = 90^\circ$,



The three ratios are defined as the functions:

$$\begin{aligned} \sin \theta &= \frac{b}{c} = \frac{\text{opposite}}{\text{hypotenuse}} & \text{SOH} &= \text{Sin Opposite Hypotenuse} \\ \cos \theta &= \frac{a}{c} = \frac{\text{adjacent}}{\text{hypotenuse}} & \text{CAH} &= \text{Cosine Adjacent Hypotenuse} \\ \tan \theta &= \frac{a}{b} = \frac{\text{opposite}}{\text{adjacent}} & \text{TOA} &= \text{Tangent Opposite Adjacent} \end{aligned}$$

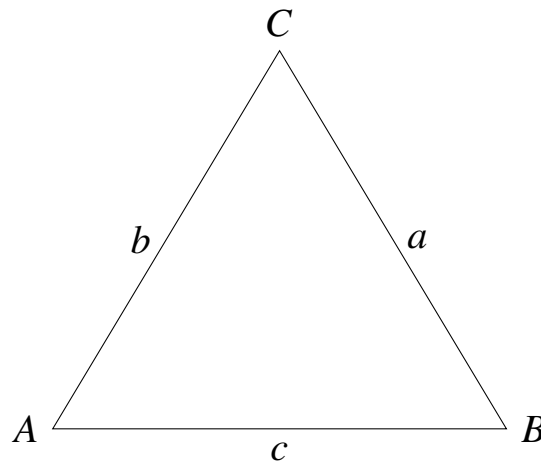
Obtuse Angles

When the 3 trigonometric functions have a Obtuse angle rather than a Acute angle, we use:

$$\begin{aligned} \sin A &= \sin O \\ \cos A &= -\cos O \\ \tan A &= -\tan O \end{aligned}$$

Where A is acute angle and O is obtuse angle. $O = 180 - A \iff A = 180 - O$ can be used to find obtuse angles when needed.





Sine Rule

Sine Rule is, ratios of side lengths to their opposite angles.

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

This is applied when one pair, a side length and its opposite angle, is given.

Cosine Rule

Cosine rule is to find the length of a side using the 2 other sides and the angle they are making.

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = c^2 + a^2 - 2ca \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

For a as unknown side, b and c as other 2 sides. $\cos A$ is angle opposite to unknown side, or angle made by sides b and c .

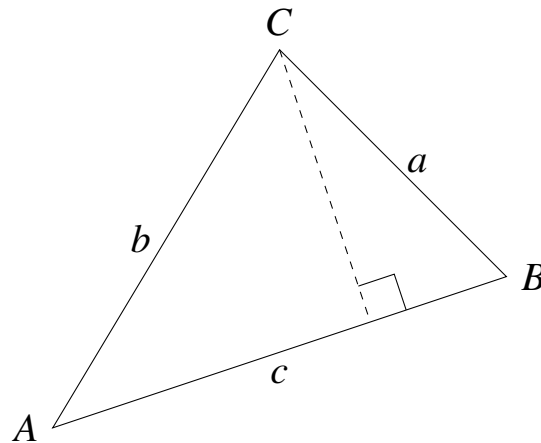
We can use this to find the angle made by b and c instead if a is given as well.

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} \quad \cos B = \frac{c^2 + a^2 - b^2}{2ca} \quad \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

All three formulas have a , b and c cycled, so same as the first, original, equation.



Area of Triangle



To find the area of a triangle which doesn't have a right angle, we use:

$$A = \frac{1}{2}ab \sin C$$

For a , b as any 2 sides, and C as the angle they make.

This is to the fact if we make a perpendicular from the base AB to C , that is the minimum distance and hence the real height. The dashed line shows this.

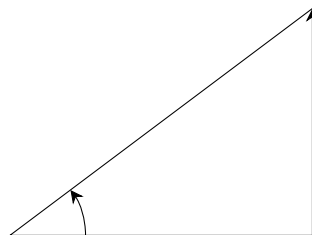
Angle of Elevation and Depression

O levels and Extended

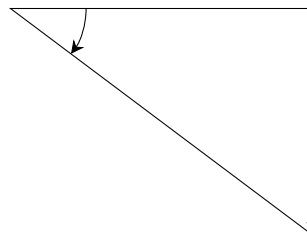
"Elevation" means height.

We measure this from a base line, e.g ground, and up to the wanted point.

Elevation



Depression



"Depression" means depth.

We measure this from a base line/high point and down to the wanted point.

Trigonometry in 3D

We need to be wary of how the diagram is, what the question demands and what/how a plane and line make a point/angle.

E.g finding the height of a pyramid, we measure from one vertex to the pyramid's apex, the angles measure from the base of the pyramid.



Trig Values and Functions

Covering

- Exact trigonometric values
- Trigonometric functions

Extended

Extended

Trigonometric values

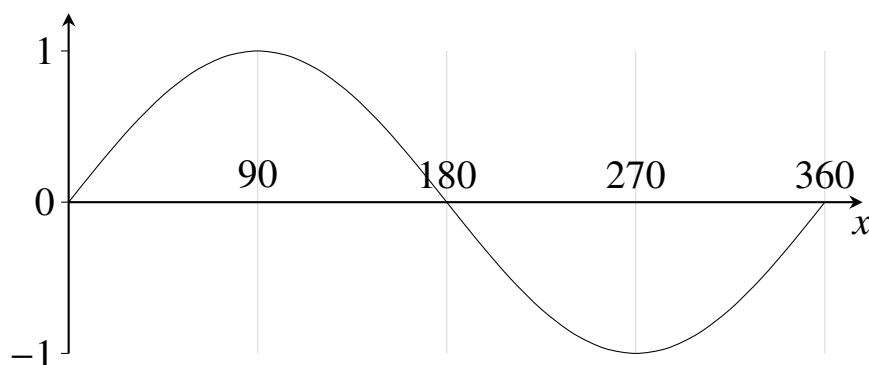
For this, a the table is good to see the values of $\sin x$, $\cos x$ and $\tan x$ for simple values of x .

x	0°	30°	45°	60°	90°
$\sin x$	$\frac{\sqrt{0}}{2}$	$\frac{\sqrt{1}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{4}}{2}$
$\cos x$	$\frac{\sqrt{4}}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{1}}{2}$	$\frac{\sqrt{0}}{2}$
$\tan x$	$\frac{\sqrt{0}}{3}$	$\frac{\sqrt{3}}{3}$	$\frac{\sqrt{9}}{3}$	$\frac{\sqrt{27}}{3}$	undefined

x	0°	30°	45°	60°	90°
$\sin x$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos x$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan x$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined

Graphs and Solution of Trig Functions

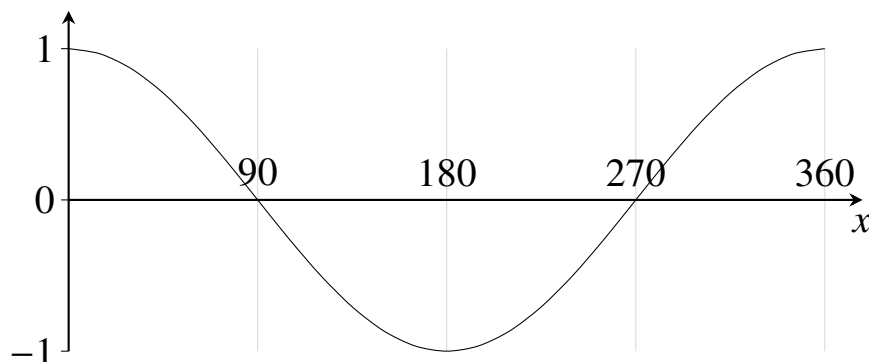
The graph of a sine function is:



To solve $\sin x = c$, we firstly have $x = \sin^{-1} c$,
but there is a second solution $x = 180 - \sin^{-1} c$.

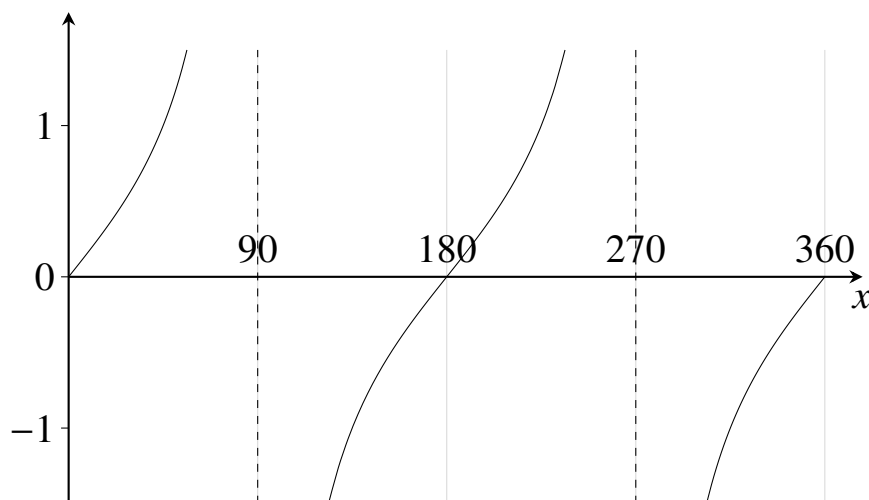


The graph of a cosine function is:



To solve $\cos x = c$, we firstly have $x = \cos^{-1} c$,
but there is a second solution $x = 360 - \cos^{-1} c$.

The graph of a tangent function is:



To solve $\tan x = c$, we firstly have $x = \tan^{-1} c$,
but there is a second solution $x = 180 + \tan^{-1} c$

